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Regular Paper

Effect of electric impulse on fruit body production of *Lentinula edodes* in the sawdust-based substrate

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Abstract

Effect of electric impulse was investigated on fruit body formation of *Lentinula edodes* in the sawdust-based substrate. Electric impulse of 15 kV was directly charged to the substrate at 3 different fruiting phases, 90- (1st flush), 130- (2nd flush), and 170- (3rd flush) day cultures, respectively. Ergosterol contents and water conditions of the substrate were examined at the 3 phases. The electric impulse was effective in fruit body formation of 4 strains for 3 different fruiting types, which occurred year-round, from autumn to winter. The effect of the electric impulse was more prominent as culture age proceeded (2nd and 3rd flushes).

Introduction

Cultivation of *Lentinula edodes* (Berk.) Pegler on the sawdust-based substrate is steadily increasing in every district. Efficient techniques are especially useful for this cultivation method[1]. Biological efficiency has been improved by optimizing various factors, such as substrate formula, strain type, culture maturity, water condition and other environmental conditions of the cultivation room[2,3].

Electric power utility has been applied to various agricultural crops, especially in horticultural fields. Cultivation systems have been improved with electricity utilization in the field of systematic house crops such as tomato, lettuce, strawberry and various kinds of flowers. Effects of air ions on plants have experimented with barley leaves[4]. Influence of the electric impulse has been tested on cultivation of some edible mushrooms. Fruit body production was promoted by the electric impulse treatment on the logwood of *L. edodes*[5,6]. We investigated the effect of the electric impulse on the fruit body formation of *L. edodes* in the sawdust-based substrate.

Materials and Methods

Strains

Four strains belonging to 3 different fruiting types were examined in this work. These were KS-1, KS-2 (for year-round cultivation), KS-6 (for cultivation in autumn) and KS-12 (for cultivation in spring). All of them are originated from a commercial source. Cultures were maintained on a potato dextrose agar (PDA, Difco) in the Petri dishes.

Culture media and growth conditions

As shown in **Fig. 1**, the substrate consisted of sawdust of *Quercus serrata* (Thunb.) et. Zucc. supplemented with wheat bran and corn powder.

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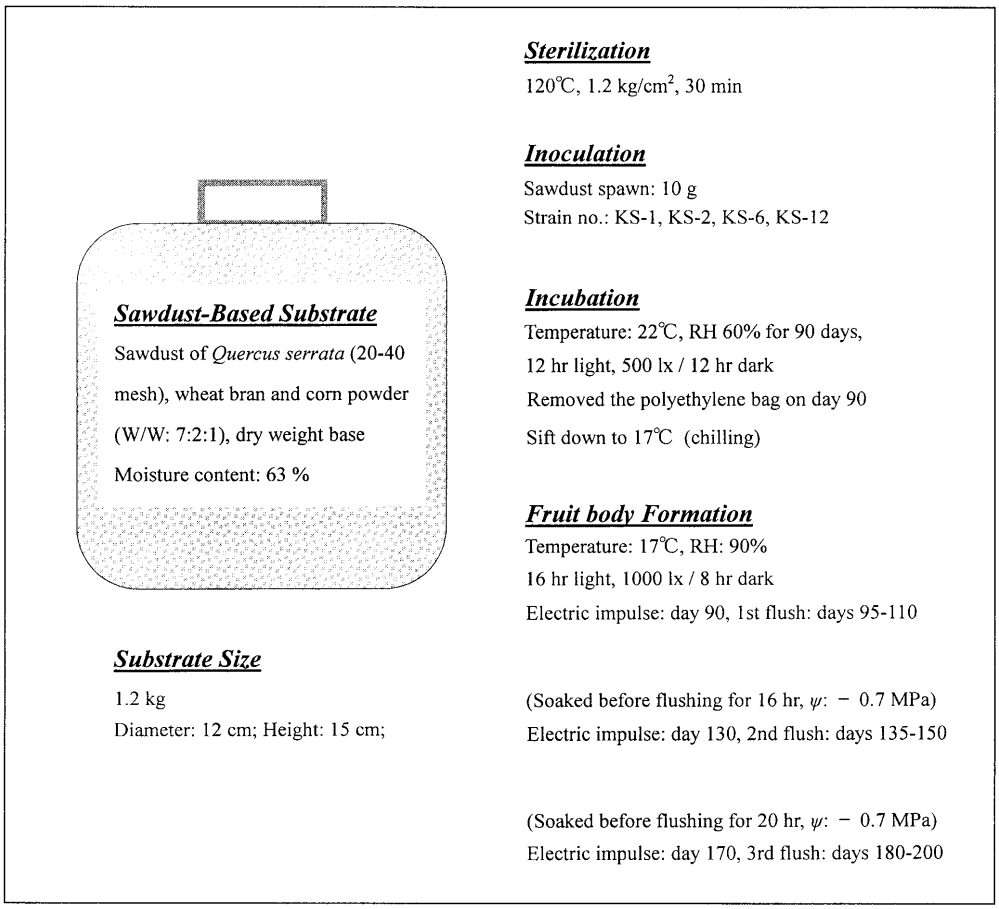

Fig. 1. Culture conditions of the sawdust-based substrate of *L. edodes*.

Moisture content was adjusted to 63 %. The mixture was packed in polypropylene bags, and was sterilized by an autoclaving. Inoculation of the substrate with spawn was done under aseptic conditions. The culture bags were kept in the incubation room at 22℃. The growth stage for the vegetative mycelium was set for 90 days. The plastic bags were completely removed before the mushroom production stage.

The colonized mature cultures were then moved to a production room. Temperature was maintained at 17℃ and relative humidity of over 90 % relative was obtained by automatic misting. Light intensity was maintained at 1000 lux by fluorescent lamps in a 16 hr light and 8 hr dark cycle per day. The substrate had three flushes in the fruiting stage. The first flush occurred spontaneously just after substrate movement. Second and 3rd flushes were induced by a soaking treatment of 16 (130-day substrate) and 20 (170-day substrate) hr, respectively. Fruit bodies were picked when the veil broke and the gills were fully exposed. All fruit bodies were counted and weighed.

Measurement of culture maturity

Ergosterol content and water potential of the substrate were measured in order to estimate the culture maturity. Ergosterol measurement was done according to the previously described metho d[7,8]. Water potential (ψ) of the cultures were measured by the thermocouple psychrometer (Wescor HR-33T micro voltmeter coupled to C-52F sample chamber)[9].

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Electric impulse treatment

Electric impulse treatments were done on day 90, 130 (after soaking) and 170 (after soaking), respectively. Electric impulse of 15 kV (1.0 kWS) was directly shot through the electrodes from the instrument (YH1G-200K-1KJC, Yamabishi Electric Co., Fukuoka, Japan) to the sawdust-based substrates (**Fig. 2**).

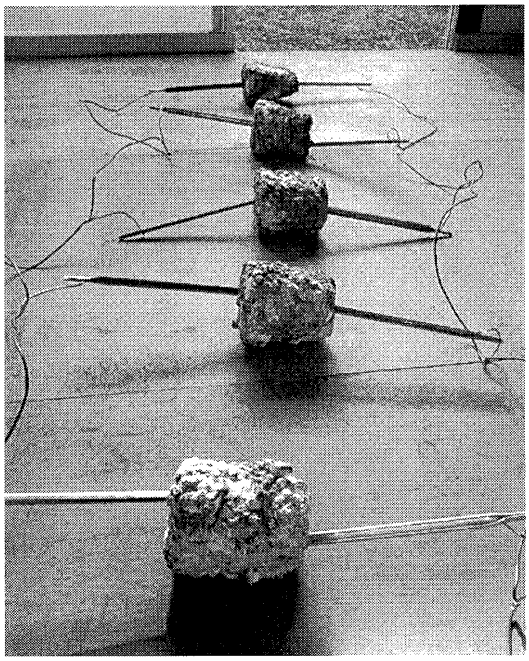

Fig. 2. Electric impulse treatment to the sawdust-based substrate of *L. edodes*.

Results and Discussion

Maturity of substrate at the electric impulse treatment

Ergosterol content

At the electric impulse treatment, the mycelial growth and physiological condition were judged from the ergosterol content and water potential. **Figure 3** shows the ergosterol content in the sawdust-based substrate. Ergosterol content is a suitable indicator for judging the substrate maturit y[1,7,8]. The substrates were moved to the production room on day 90. First, ergosterol content was measured at this phase. Ergosterol content of the

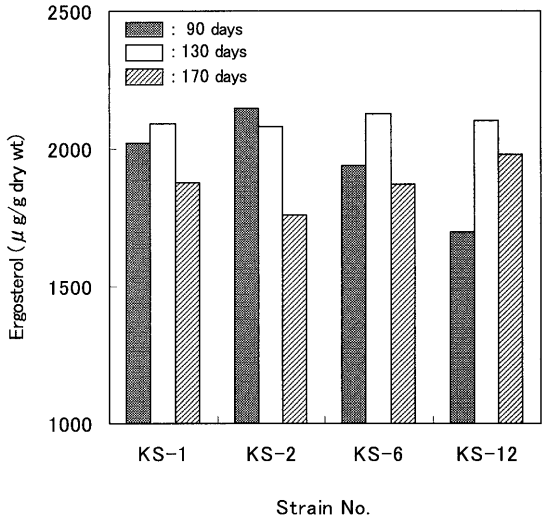

Fig. 3. Ergosterol content of the sawdust-based substrate of *L. edodes* at different incubation days.

KS-6 strain was high, followed by the KS-1 and KS-2 strains. The low level of ergosterol content in the KS-12 strain suggests insufficient maturation of the sawdust-based substrate. The full grown substrate in the ergosterol content was more than 2000 $\mu g/g$[1,7]. The second flush was induced on day 130 by the water soaking treatment. Ergosterol content of all substrates is 2000 $\mu g/g$ at this stage. The third flush was induced on day 170 by the water soaking treatment. Ergosterol content was decreased because of the fruit body formation during the first and second production stages. The substrate of the strains for autumn and year-round cultivation seemed to have matured fully. Only the strain for winter cultivation maintained useful culture maturity at this stage.

Water potential (ψ)

Water condition of the sawdust-based substrate is one of the most important factors on the fruit body formation[9,10]. The ψ of the substrate has a clear relationship to the capacity of fruit body formation. The value of ψ for spontaneous fruiting was from 0.5 to 0.7 MPa[9]. The substrates of three strains except for the KS-12 strain were suitable

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for fruit body formation on the 90th day (**Fig. 4**). The KS-12 strain has not reached sufficient maturation at day 90 under the culture conditions. On day 130, all of the substrates were suitable for fruit body formation. The values of ψ were about 0.5 MPa at the growth phase. On day 170, the values of ψ were quite high because of the senescence of the substrates. At the growth phase the capacity of water holding might be decreased. The KS-12 strain maintained the suitable water condition for fruiting even on day 170.

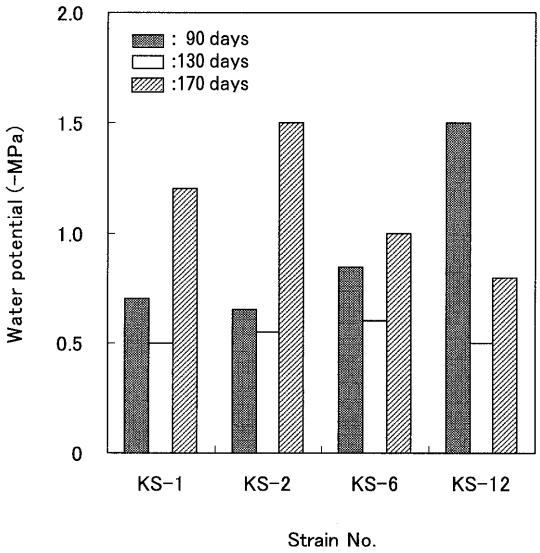

Fig. 4. Water potential of the sawdust-based substrate of *L. edodes* at different incubation days.

Effect of electric impulse on fruit body production

The first flush spontaneously occurred at 5 to 15 days after the electric impulse treatment. **Figure 5** shows the yield of fruit bodies during the 1st flush period. Electric impulse resulted in more productions of fruit-bodies in all tested strains. A difference between the electric impulse and control was already recognized in the primodia formation stage. The number of primodia on the treated substrate was more than that on the control. The normal shapes for the fruit body in the treated substrates were pilei and stipe.

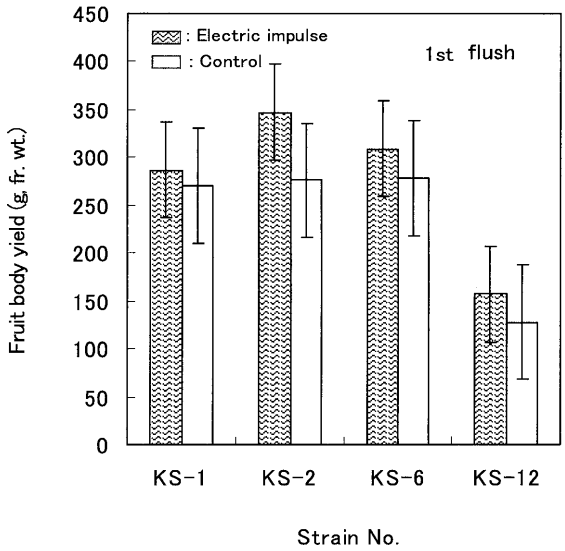

Fig. 5. Effect of electric impulse at 1st flush stage. Vertical bars indicate the standard errors of the mean (n = 30).

The second flush was induced by water soaking for 16 hr on day 130. Fruit body formation occurred at 5 to 20 days after the electric impulse treatment. **Figure 6** shows the yield of a fruit body during the 2nd flush period. The effects appeared more clearly in the 2nd flush treatment than in the 1st flush period.

The third flush was also induced by water soaking for 20 hr on day 170. Fruit body formation occurred at 10 to 30 days after the electric impulse treatment. **Figure 7** shows the yield of a fruit body during the 3rd flush period. In the 3rd flush period the effect of the electric impulse was the strongest of all periods. The fruiting capacity was actually promoted by the electric impulse. A decrease of fruit body production in the 3rd flush is the usual pattern on the sawdust-based substrate of *L. edodes*, but the electric impulse actually improved the yield.

The electric impulse treatment promoted the fruit body formation of *L. edodes*. The effect was clearer on the aged substrates at the 3rd flush. The electric impulse treatment will contribute to

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the sawdust-based cultivation of *L. edodes*. We have also reported the possibility of enhancement of fruit body formation of the mycorrhizal fungi (*Laccaria laccata*) with electric impulses in nurseries and plantations[11]. Further works are needed to clarify the relationship between fungal physiology and the electric impulse.

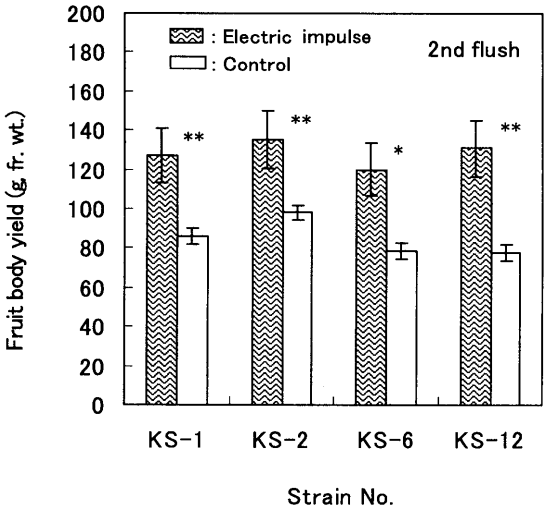

Fig. 6. Effect of electric impulse at 2nd flush stage. Vertical bars indicate the standard errors of the mean (n = 30). Asterisks indicate the significant differences at $p < 0.01$ (**) and 0.05 (*), respectively.

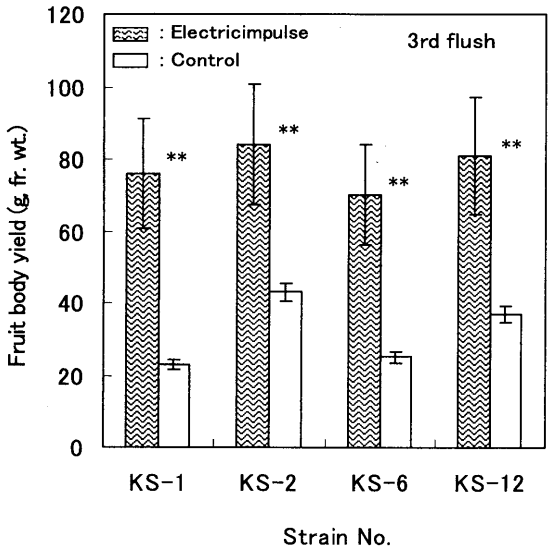

Fig. 7. Effect of electric impulse at 3rd flush stage. Vertical bars indicate the standard errors of the mean (n = 30). Asterisks indicate the significant differences at $p < 0.01$.

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和 文 摘 要

シイタケ菌床での子実体発生におよぼす電気インパルスの影響

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シイタケ菌床栽培での子実体発生におよぼす電気インパルスの効果について検討した．培養日数が 90（初回発生），130（2回目発生），170（3回目発生）の菌床それぞれに 15 kV の電気インパルスを印加した．菌床の状態を把握するために，印加する際のエルゴステロール含有量と水ポテンシャルを測定した．本研究で試験した子実体発生型が異なる 4 品種とも，電気インパルス印加によって子実体発生量が増大する効果が認められた．その効果は，培養齢が進んだ 2 回目，3 回目発生においてより顕著に表れた．

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